# WiFi Deadzone Heatmap — Antigravity Handoff & Build Plan

**Audience:** an AI c ding agent (Antigravity / Andr id Studi assistant) w rking inside a cl ne f , plus the human running it. **Goal:** add ONE new tab, **Heatmap**, t the existing app, keep every existing feature, and pr duce a w rking, installable APK. **Deadline context:** semester pr ject, ~1 day. Prefer the simplest thing that w rks reliably ver clever things.

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## 0. Files you are given

| File | Purpose | Where it goes |
|---|---|---|
| The wifianalyzer rep (cl ned) | Base app | Pr ject r t |
| ("the SPEC") | Draft K tlin c de, XML lay ut, checkp int c rdinate table | C py int rep at f r reference. D NOT leave it in |
| Fl r plan images (PNG/JPG) | Ind r maps | Put in first; the agent m ves them t with c rrect names (see §6) |
| This file | Requirements + fixes + rder f w rk | |

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## 1. Non-negotiable rules

1. **Do not remove or disable any existing feature, tab, setting, or screen** f wifianalyzer. Heatmap is purely additive, as the **last** navigati n item.
2. **Do not raise** , r upgrade Gradle/AGP/K tlin, unless the pr ject will n t build with ut it. Match the existing language, architecture and c de style f the rep .
3. **No new heavy dependencies.** Use nly what the pr ject already has (Andr idX, Material). If a Material class (e.g. ) is needed and Material is absent, use instead.
4. **No database / no network / no cloud.** Persistence is + JSON nly.
5. **Never crash.** Every failure path (Wi-Fi ff, permissi n missing, n t c nnected, missing image, empty hist ry) must sh w a friendly message and leave the app usable.
6. Keep the riginal **GPL-3.0 license** and n tices intact (wifianalyzer is GPL-3.0; derivative w rk must stay GPL-c mpatible). Add a credit line in the Ab ut screen nly if trivial; therwise skip.
7. **Run the existing unit tests.** Adding a nav item may break tests that assert menu size r rder. Update th se tests t include the new item; d n t delete tests.

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## 2. Feature requirements (from the project owner)

### 2.1 Where it lives

- New navigati n item **"Heatmap"**, placed **last** in the navigati n list. Opens .
- Existing tabs keep their rder and behavi r. On app launch the app still pens n the Access P ints list as t day.

### 2.2 Screen layout



- **Top-left: History** butt n → b tt m sheet f saved heatmaps f r the current fl r.
- **Top-right: [+]** butt n → fl r ch ser (list dial g: fl r display names). Ch sing a fl r l ads its map. The current fl r name is sh wn in the t p bar centre.

- **RSSI badge** sh ws the live signal f the c nnected netw rk (e.g. ), r if unavailable.
- **Floor chooser replaces the Spinner** in the SPEC lay ut. Default fl r n first pen: first fl r in the list.
- If a survey is in pr gress r unsaved heatmap exists and the user changes fl r, **confirm** ("Discard current survey?") bef re switching.

### 2.3 Floors (hard-coded)

- Fl r plans and checkp ints are **hard-coded** in . N user-upl aded maps.
- The list f fl rs is **data-driven**: adding a fl r = ne new entry plus ne image. N ther c de changes.
- The SPEC includes 4 fl rs: CSE, ENTC/EXTC gr und, ENTC/EXTC first, Instrumentati n. The wner may als supply an **IT department** image. **Use whatever images are actually provided** (see §6 f r h w t handle a fl r with n c rdinates).
- Blackened areas (stairs, pen-t -sky v ids, t ilets) are part f the image; checkp ints must nly be placed n usable areas (c rrid rs, r ms).

### 2.4 Survey state machine

| State | Visible buttons | Rules |
|---|---|---|
| **IDLE** | | Checkp ints drawn as h ll w circles; taps ign red. Hist ry available. |
| **SURVEYING** | ; c unter | Tapping a checkp int rec rds RSSI. First tap: t ast . Re-tap: **overwrite** and t ast . |
| **STOPPED** | , , , | is **disabled until ≥ 3 checkpoints** are sampled (sh w in the label r a hint). |
| **HEATMAP** | , | Heatmap drawn fr m sampled checkp ints nly. writes t hist ry and sh ws a t ast; the heatmap stays n screen (Save butt n then sh ws "Saved ✓" and is disabled). returns t IDLE. |
| **HISTORY VIEW** | nly | Opened fr m a hist ry entry; read- nly heatmap. |

- Tapped checkpoint **flashes yellow/pulses briefly (~700 ms)** then becomes a **solid dot colored by its RSSI**. A simple ValueAnimator wrapped/invalidate is enough; don't over-engineer.
- Show the checkpoint **name** in the toast; on the canvas, draw a short label (ID number, or name if it fits) so labels don't overlap into clutter. Suggestion: ID inside/next to circle, name in toast.
- A checkpoint tap needs a valid RSSI. If none is available, **do not record**; show .

### 2.5 Heatmap rendering

- Normalise RSSI to 0.0–1.0 and map to hue 0° (red) → 120° (green): , .
- Constants: , (practical indoor range; SPEC used -100/-35 — either is acceptable, but keep them as two named constants so they are easy to change).
- Each sampled checkpoint draws a **soft radial gradient** (fixed influence radius per checkpoint). Unsampled areas stay **uncolored / neutral**. Never interpolate invented data into unsampled corridors.
- Splats are semi-transparent (alpha ≈ 150–180) and overlap naturally.
- Heatmap generates **only** from collected checkpoint data.

#### 2.5.1 History (local storage)

- Stored in as JSON, one key per floor: → array of records:

```
json
```

- **Cap at 20 saved heatmaps per floor** (drop the oldest) so storage never bloats.
- History bottom sheet: newest first; each row shows date/time, SSID, and point count. **Tap** = open that heatmap. **Long-press or a delete icon** = delete that entry (with confirm).

- When pening a saved entry, **render from the stored** `normX/normY`, n t fr m the current rep sit ry c rdinates, s recalibrating checkp ints later d es n t dist rt ld surveys.
- Empty hist ry → t ast .
- All JSON parsing must be wrapped in try/catch; a c rrupt entry is skipped, never crashes.

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## 3. Existing-app integration (agent must discover, not assume)

wifianalyzer's navigati n is defined in a few places (typically a navigati n-menu enum, a menu XML such as , and /a navigati n helper that maps items t fragments). The exact names vary by versi n. **Inspect the repo first**, then:

1. Find h w existing items (Access P ints, Channel Rating, Channel Graph, Time Graph, Exp rt, Channel Available, Vend rs, P rt Auth rity, Settings, Ab ut…) are declared, rdered, given ic ns/titles, and mapped t fragments.
2. Add **Heatmap** the same way, **last in order**, with a built-in ic n (e.g. ) r a vect r drawable.
3. If navigati n is a **bottom bar with a 5-item cap**, d n t rem ve tabs; keep using whatever verfl w/drawer mechanism the app already has and place Heatmap there. If it is a drawer, add it at the b tt m f the main gr up.
4. Update any **unit tests** that enumerate navigati n items.
5. Add all new user-visible text t . If lint c mplains ab ut missing translati ns ( ), suppress it f r the new strings ( ) instead f editing every l cale file.
6. L k at h w the app already gets Wi-Fi/scan data (e.g. an existing scanner service / c nnecti n bject exp sing the c nnected netw rk's signal level). **If** it cleanly exp ses the c nnected AP's RSSI, reuse it (it already handles scan scheduling and permissi ns). **Otherwise** use as in §4. Pick ne appr ach; d n't mix.
7. Permissi ns: the app already requests l cati n permissi n f r scanning. Verify , , and are in the manifest; add nly what is missing. The Heatmap screen must handle "permissi n n t granted" and "L cati n services ff" gracefully (message, n crash).

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## 4. RSSI reading rules

- Wi-Fi scan thr ttling n Andr id 9+ limits . **The owner will disable "Wi-Fi scan throttling" in Developer Options**, s the app can scan frequently. The c de must still **work (more slowly) with throttling on**, never crash.

- While the Heatmap fragment is resumed, run a lightweight **poller every ~1.5–2 s** (a /c r utine tied t the fragment lifecycle; st p it in ) that: (a) pti nally calls inside try/catch (deprecated n newer APIs but still w rks), (b) reads the c nnected BSSID's level fr m if present, else , (c) updates the badge and caches the latest value.
- **Invalid values:** treat , , r any value as "n signal". Treat / null BSSID as "n t c nnected".
- On checkp int tap, rec rd the **latest cached valid RSSI**. (Opti nal impr vement if time permits: average the last 3 samples.)
- is deprecated n API 31+; **suppress the deprecation warning** ( ) rather than rewriting f r . It still w rks.
- SSID: strip qu tes; if the value is (n permissi n), st re .

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## 5. Mandatory fixes to the SPEC code

The SPEC c de is a g d base, but apply all f the f ll wing:

1. **Overwrite toast.** In the checkp int tap handler: if sh w , else .
2. **Start / Stop flow.** Replace the SPEC's "Cancel Survey" with the state machine in §2.4. Implement it as an explicit and a single functi n that sets butt n visibility/text. D n t scatter visibility changes acr ss click listeners.
3. **Gradient fade colour.** Replace with (same c l ur, alpha 0) t av id dark/grey muddy edges.
4. **Relative radius.** Replace fixed px with a value derived fr m the drawn map, e.g. (within the 150–200 px spirit n a typical ph ne). Same f r t uch radius (use → px) and circle/text sizes (use / via ), s it l ks right n every ph ne.
5. **No allocation in `onDraw`.** The SPEC creates and bjects per checkp int per frame. Cache ne ; create/update shaders nly when readings r size change ( r reuse a Paint and set per p int, but d n t c nstruct new Paints in the l p).
6. **Bitmap safety.**
   - Put images in s Andr id d esn't rescale them.

- Dec de with [redacted] ([redacted] s the l ngest side ≈ 2000 px r less). Never recycle a bitmap that [redacted] might still use: rem ve the manual [redacted] call r guard [redacted] with [redacted].
- If dec ding returns null, sh w a t ast and return; d n t crash.

7. **Lifecycle & rotation.** Keep readings and state acr ss c nfigurati n changes (a small [redacted] r [redacted] with the readings map and state). At minimum, r tati n r app backgr unding must n t silently wipe a survey in pr gress. (Opti nal: l ck this screen t p rtrait if state rest rati n pr ves flaky, but nly as a last res rt.)
8. **Layout compatibility.** Use [redacted] instead f [redacted]/ [redacted] if the pr ject's [redacted] < 26. Use [redacted] r [redacted] c nsistent with the pr ject's Andr idX setup instead f [redacted] if lint c mplains. Use the pr ject's existing theme/c l rs where p ssible; a dark l k is fine.
9. **Replace the Spinner with the `[+]` floor chooser** (§2.2). Rem ve the Spinner-initial-selecti n side effects (the SPEC's [redacted] calls [redacted] n first sh w).
10. **History hardening.** Cap at 20 entries, st re [redacted], newest-first listing, delete supp rt, try/catch ar und every JSON read, apply with [redacted].
11. **History restore uses stored coordinates** (§2.5.1). The canvas sh uld be able t render a list f [redacted] samples independent f [redacted] checkp ints.
12. **Legend.** Draw a small h riz ntal gradient bar (red→green) with [redacted] / [redacted] ( r the dBm values) in a c rner f the canvas r just ab ve the b tt m bar.
13. **Hit-testing.** Taps are tested against checkp ints using the [redacted] mapping; ign re taps utside [redacted]. If tw checkp ints are within the t uch radius, ch se the **nearest**.
14. **Do not draw heatmap over a never-sampled floor.** Heatmap butt n is the nly way t turn it n, and it requires ⩾ 3 samples.
15. **Optional polish (skip if short on time):** clip splats s c l ur d esn't bleed nt pure-black (blackened) pixels; a subtle pulse animati n instead f a ne-time flash.

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## 6. Assets and checkpoints

### 6.1 Image naming and mapping

M ve images fr m [redacted] t [redacted] using **lowercase, no spaces, letters/digits/underscore only**, mapped as:

| Likely source | Resource name | Floor ID (from SPEC) |
|---|---|---|
| CSE department | | |
| EXTC/ENTC gr und | | |
| EXTC/ENTC first | | |
| Instrumentati n gr und | | |
| IT department (if pr vided) | | |

If the file names d n't clearly say which building/fl r they are, **look at the images and infer from room labels**; if still unclear, list y ur mapping in y ur final rep rt s the human can c rrect it.

### 6.2 Floors with no coordinates (e.g. IT)

The SPEC has n checkp int table f r IT. If an IT image is pr vided:

- Add a with **8–12 provisional checkpoints** placed n c rrid r/r m centres y u can identify in the image (n rmalised 0–1, relative t the full image), named fr m visible r m labels where readable.
- Mark them with a c mment and **list them in the final report**. If n IT image is pr vided, d n t create IT.

### 6.3 Coordinates are estimates, so include a debug calibration mode

The SPEC's c rdinates are appr ximati ns. Add a **debug-only** calibrati n aid s the human can fix them in minutes:

- L ng-press anywhere n the canvas ( nly when ) → t ast and .
- It must be **inactive in release builds**. The human then edits values. (The agent may als d a first visual pass t nudge any checkp int that clearly lies n a blackened area r utside the building.)

### 6.4 Display names

Use readable names, e.g. "CSE Department", "EXTC Gr und Fl r", "EXTC First Fl r", "Instrumentati n Dept", "IT Department". Keep IDs stable ( etc.) since they are st rage keys. The wner may rename display names nly.

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## 7. Work order for the agent (do one phase, build, then continue)

**Phase A — Baseline (no changes).** Open the pr ject, sync Gradle, build the debug APK unm dified. Rep rt JDK/Gradle pr blems and fix **only** what's needed t build. Run existing unit tests and rec rd the baseline result.

**Phase B — Inspect.** Read navigati n c de (§3), minSdk/targetSdk, existing scanner/permissi n c de. Write a sh rt summary (≤ 15 lines): navigati n files f und, h w t add an item, minSdk, whether the existing scanner exp ses c nnected RSSI.

**Phase C — Add feature code.** Create package [ ] (adapt t the pr ject's actual package/namespace) c ntaining: [ ]/[ ], [ ], [ ], [ ], [ ] (+ pti nal [ ]), lay ut [ ], strings. Start fr m the SPEC and apply every fix in §5. M ve images (§6). Build until it c mpiles.

**Phase D — Wire the tab.** Add the Heatmap navigati n item (§3). Build, run all unit tests, fix/update tests that enumerate navigati n items.

**Phase E — Self-review.** Re-read all new files l king f r: crashes n null WifiManager, disc nnected Wi-Fi, missing permissi n, empty hist ry, r tati n, bitmap recycling, ff-by- ne in "n/3", and state/visibility bugs. Fix them.

**Phase F — Build APK.** Run [ ] and rep rt the path [ ]. Run [ ] ( r the pr ject's lint task) and fix any **new** err rs y u intr duced (d n't mass-fix pre-existing warnings).

**Final report (required):** list f files created/changed, the navigati n mapping decisi ns, fl r-image mapping, pr visi nal checkp ints needing calibrati n, anything y u c uld n t verify, and exact build c mmands used.

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## 8. Test checklist (human, on a real phone)

Pre-reqs: Devel per pti ns → Netw rking → **Wi-Fi scan throttling OFF**; Wi-Fi c nnected t the target netw rk; L cati n ON; app granted L cati n permissi n.

1. All riginal tabs still pen and w rk (Access P ints list, graphs, Exp rt, Settings, Ab ut, etc.).
2. Heatmap tab appears last; pens with ut crashing.
3. [ ] lists all fl rs; ch sing ne sh ws its map c rrectly (n t stretched; checkp ints n the right places).
4. RSSI badge updates as y u walk (value changes within a few sec nds).
5. Start → tap a checkp int: yell w flash → s lid c l ured d t; t ast sh ws name and dBm.

6. Re-tap same checkp int: t ast ; value changes.
7. C unter c rrect; St p sh ws Generate disabled bel w 3 p ints, enabled at 3+.
8. Generate: s ft red/yell w/green splats; unsampled areas neutral; legend visible.
9. Save → t ast; pen Hist ry → entry with date/time, SSID, p int c unt; tapping it redraws the same heatmap.
10. Delete a hist ry entry; re pen t c nfirm it's g ne.
11. R tate the ph ne mid-survey: data n t l st.
12. Turn Wi-Fi ff / disc nnect: tapping a checkp int sh ws the friendly message, n crash.
13. Switch fl rs with an unsaved survey: c nfirmati n appears.
14. Kill and re pen the app: hist ry persists.
15. Install the APK n a sec nd ph ne and repeat steps 2–9.

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## 9. Human-only tasks (cannot be done by the AI)

1. **Prepare images**: PNG, ab ut 1080–2000 px n the l ng side, stairs/t ilets/v ids blackened, high c ntrast walls.
2. **Calibrate coordinates on site** with the debug l ng-press helper (§6.3): stand at/see each checkp int, l ng-press the same sp t n the map, c py fr m L gcat, update .
3. **Confirm naming**: EXTC vs ENTC, "gr und/first" vs "first/sec nd", and whether IT is included.
4. **Collect demo data**: save at least ne survey per fl r bef re presenting s Hist ry isn't empty.
5. **Take screenshots** (red-t -green heatmaps, hist ry sheet) f r the rep rt.

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## 10. Building and sharing the APK

- **Quick share (fine for a semester project):** → . Send the file; recipients must all w "Install unkn wn apps".
- **Release build (optional):** Andr id Studi → Build → Generate Signed App Bundle/APK → APK → create a keyst re (st re it and its passw rds safely) → release. If R8/minify breaks the app, the agent sh uld n t enable m re aggressive shrinking; keep the pr ject's existing settings.
- Opti nal branding: change the app **label** (display name) nly. Changing is all wed if the wner wants it t install beside the riginal app, but d it nly in , never rename s urce packages.

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## 11. Copy-paste prompts for Antigravity (in order)

### Prompt 1 — baseline + inspect

> This is a cl ne f wifianalyzer. Read fully. D Phase A and Phase B nly: build the unm dified app, run existing tests, then summarize h w navigati n is defined, minSdk/targetSdk, and whether the existing scanner exp ses the c nnected netw rk's RSSI. D n t change any files.

### Prompt 2 — feature code

> D Phase C fr m the hand ff d cument. Use as the starting c de and apply every fix in secti n 5 (hand ff wins n c nflicts). Images are in ; map them per secti n 6. D n't t uch navigati n yet. Build until it c mpiles and tell me what y u changed.

### Prompt 3 — wire the tab

> D Phase D: add Heatmap as the last navigati n item per secti n 3, with ut rem ving r disabling any existing feature. Update unit tests that c unt r list navigati n items. Build, run tests, and sh w me every file changed.

### Prompt 4 — review and APK

> D Phases E and F: self-review the new c de against the checklist, fix issues, run lint/tests, build the debug APK, and give me the final rep rt described in secti n 7.

**If something fails:** paste the full err r text and the file it p ints t . If an err r c mes fr m the existing c de, fix it minimally and tell me; d n't refact r.

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## 12. Definition of done

- App builds, installs, launches; all riginal features w rk; unit tests pass.
- Heatmap is the last tab; fl r selecti n via ; Start/St p/Generate/Save/Clear fl w per §2.4.
- Overwrite t ast, ≥ 3 checkp int rule, 20-entry hist ry cap, r tati n-safe state, graceful n -Wi-Fi/permissi n handling.
- Heatmap visually c rrect n a real ph ne with the calibrated checkp ints; at least ne saved dem survey per fl r.
- A signed r debug APK is pr duced and installs n a sec nd device.